TCP va UDP asoslari

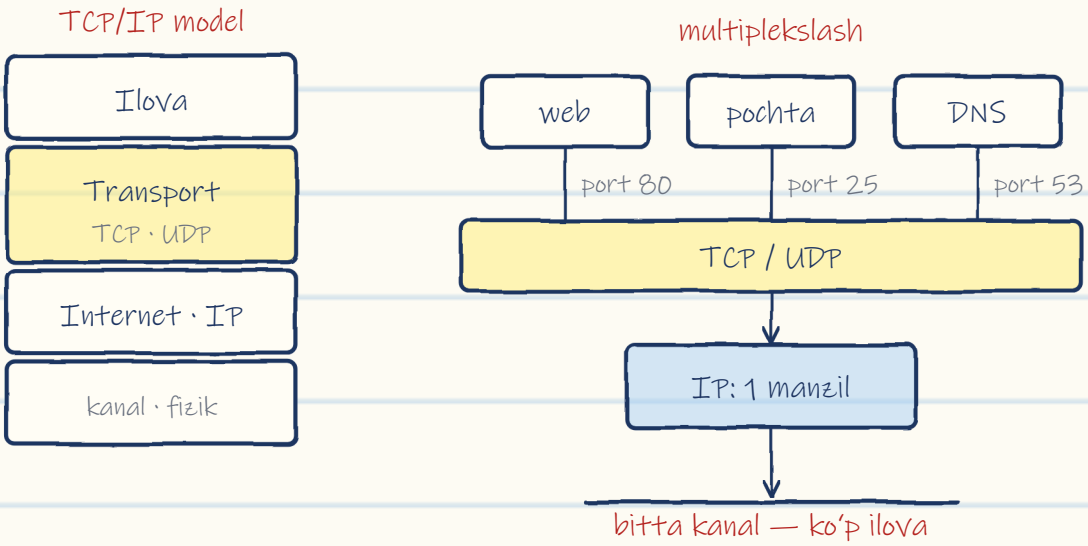
TCP and UDP Fundamentals

Asosiy atamalar
- TCP — ulanishli, ishonchli transport protokoli (RFC 793)
- UDP — ulanishsiz, tasdiqsiz, tez protokol (RFC 768)
- Port — bitta xostdagi ilovalarni ajratuvchi raqam
- Segment — transport qatlami uzatadigan ma'lumot bo'lagi

1. Transport qatlami vazifalari

Ilova va IP orasida — xostdan xostga ulanish.

- ulanishli / ulanishsiz xizmat
- holatni saqlash — ulanishlar bazasi
- segmentlash — oqimni bo'laklarga bo'lish
- multiplekslash — port bilan ilovani ajratish
- ishonchlilik — faqat TCP
- oqim nazorati — tezlikni moslash



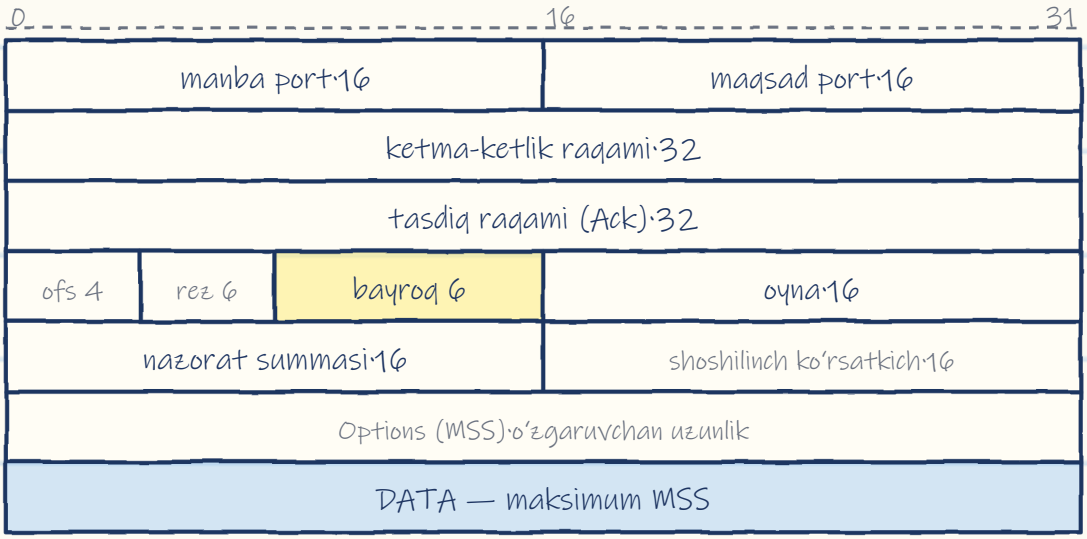
1-rasm. Bir IP manzil — ko'p ilova, portlar ajratadi

Manba: TCP — RFC 793, UDP — RFC 768 — ikkisi IP ustida.

2. TCP xususiyatlari

- uch qadamli qo'l siqish
- port raqami — ilovani aniqlaydi
- nazorat summasi — xatoni topadi
- tasdiq (Ack) — kelmasa qayta uzatiladi
- ketma-ketlik raqami — tartib va yo'qolish
- siljuvchi oyna — oqim nazorati

3. TCP segment sarlavhasi

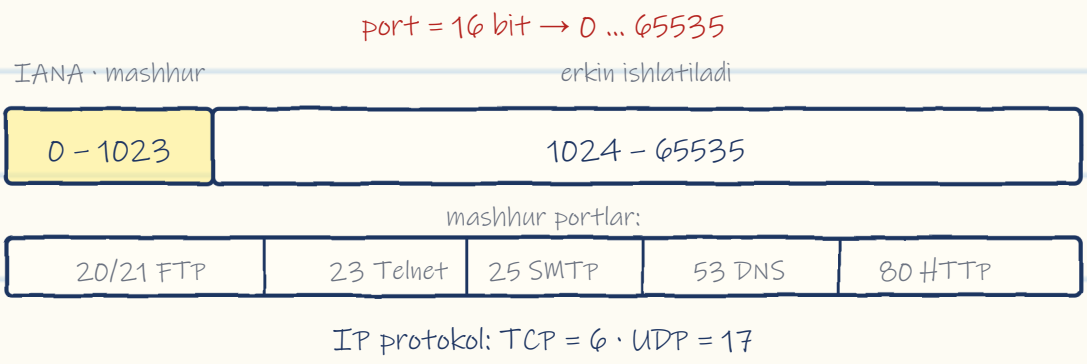


2-rasm. TCP sarlavhasi — kamida 20 bayt

Bayroqlar: URG, ACK, PSH, RST, SYN, FIN

MSS: odatda 1024 B, Ethernet — 1460 B; ikki tomon kelishadi.

4. Port raqamlari

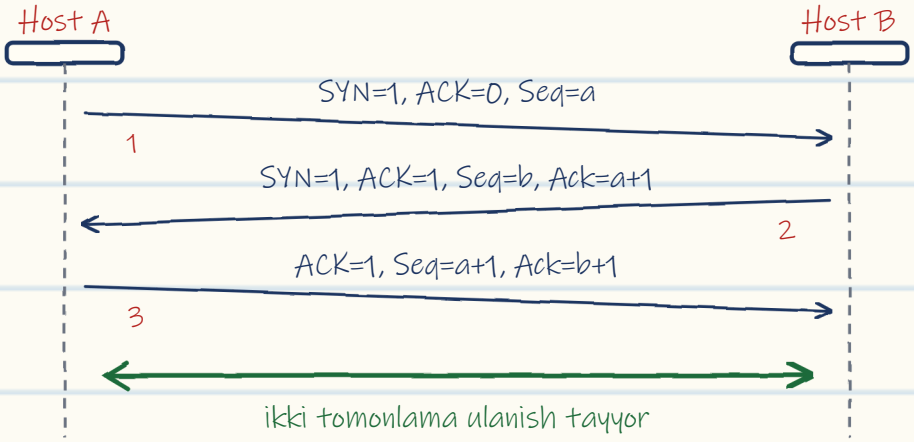


IP protokol: TCP = 6 · UDP = 17

3-rasm. 0–1023 — IANA (RFC 1700), qolgani bo'sh

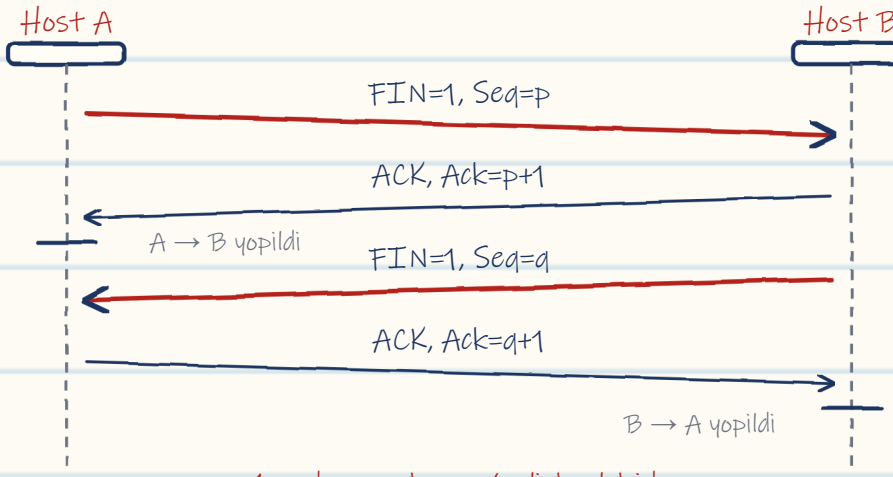
Ulanishni aniqlaydi	manba IP · manba port · maqsad IP · maqsad port
Qayerda	IP — IP sarlavhada, port — TCP/UDP sarlavhada

5. Ulanish o'rnatish — 3 qadam



4-rasm. SYN — qo'l siqish belgisi

6. Ulanishni yopish — 4 qadam

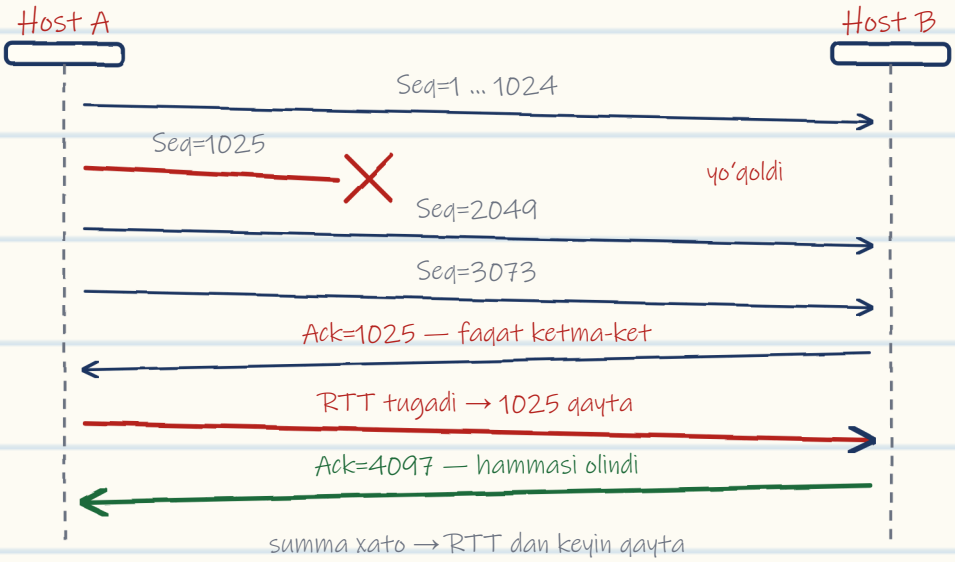


4 qadam — har yo'nalish alohida

5-rasm. FIN — yopish belgisi

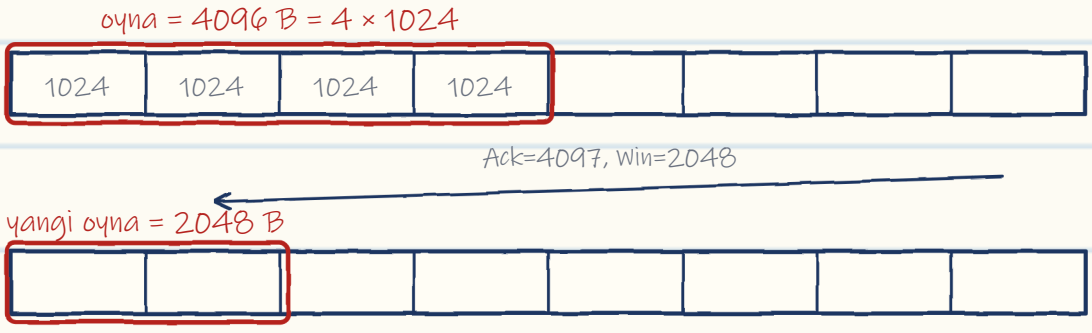
7. Ishonchlilik: tasdiq va qayta uzatish

- Ack=N+1 — N dan kichik hammasi olindi
- tasdiq ma'lumot bilan birga ketadi
- RTT tugagach qayta uzatiladi



8. Siljuvchi oyna

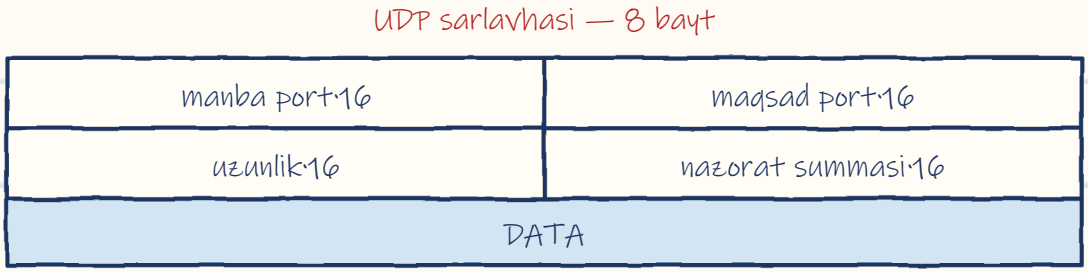
Oyna = bir marta kutiladigan bayt soni; 16 bit → 65535.



6-rasm. B buferi kam — A tezlikni yarmiga pasaytiradi

Cheklov: oyna faqat xostlar orasida — router tiqilinchini TCP ko'rmaydi.

9. UDP sarlavhasi



ulanishsiz · tasdiq yo'q · qayta uzatish yo'q

7-rasm. Uzunlik = sarlavha + ma'lumot

10. TCP va UDP — taqqoslash

	TCP	UDP
ulanish o'rnatish	✓	✗
tasdiq va qayta uzatish	✓	✗
siljuvchi oyna	✓	✗
sarlavha	20+ bayt	8 bayt

UDP: VoIP · video · BOOTP — kechikish kichik

8-rasm. UDP ishonchli tarmoq uchun

UDP afzalligi: sodda, holat va bufer yo'q — polosani tejaydi.

Xulosa

- Transport qatlami: port · segment · oqim · ishonchlilik
- Port 16 bit; 0–1023 mashhur; TCP=6, UDP=17
- Ochish — 3 qadam (SYN), yopish — 4 qadam (FIN)
- Seq va Ack: tartib, yo'qolish, takror
- Oyna ≤ 65535 bayt; Win=0 — to'xtash
- UDP: 8 bayt, tasdiq yo'q, kechikish kichik

Tekshirish

1. TCP sarlavhasi minimal uzunligi? — 20 bayt
2. Ethernet uchun MSS? — 1460 bayt
3. Mashhur portlar diapazoni? — 0 … 1023
4. Oyna maksimumi va Win=0 ma'nosi? — 65535 B; to'xta
5. UDP sarlavhasida nechta maydon? — 4 ta